

Lab # 11

Mutability and Immutability

Lab Tasks:-

1. Apply concept of mutability and immutability for the task promoted and failed students. The roll number, subject code, and subject name would have to be entered at time of object creation and with getter method these variables should be printed. (Hint: Those students who are failed in previous semester will be registered in immutable class, and promoted students are registered in mutable class)

Code:-

```
FailedStudent.java x PromotedStudent.java StudentTest.java
1 final class FailedStudent { no usages
2     private final int rollNumber; 2 usages
3     private final String subjectCode; 2 usages
4     private final String subjectName; 2 usages
5
6     public FailedStudent(int rollNumber, String subjectCode, String subjectName) {
7         this.rollNumber = rollNumber;
8         this.subjectCode = subjectCode;
9         this.subjectName = subjectName;
10    }
11
12    public int getRollNumber() { no usages
13        return rollNumber;
14    }
15    public String getSubjectCode() { no usages
16        return subjectCode;
17    }
18    public String getSubjectName() { no usages
19        return subjectName;
20    }
21 }

FailedStudent.java PromotedStudent.java x StudentTest.java
1 class PromotedStudent { no usages
2     private int rollNumber; 3 usages
3     private String subjectCode; 3 usages
4     private String subjectName; 3 usages
5
6     public PromotedStudent(int rollNumber, String subjectCode, String subjectName) {
7         this.rollNumber = rollNumber;
8         this.subjectCode = subjectCode;
9         this.subjectName = subjectName;
10    }
11    public int getRollNumber() { no usages
12        return rollNumber;
13    }
14    public String getSubjectCode() { no usages
15        return subjectCode;
16    }
17    public String getSubjectName() { no usages
18        return subjectName;
19    }
20    public void setRollNumber(int rollNumber) { no usages
21        this.rollNumber = rollNumber;
22    }
23    public void setSubjectCode(String subjectCode) { no usages
24        this.subjectCode = subjectCode;
25    }
26    public void setSubjectName(String subjectName) { no usages
27        this.subjectName = subjectName;
28    }
```

```

FailedStudent.java  PromotedStudent.java  StudentTest.java x
1  public class StudentTest {
2  public static void main(String[] args) {
3      FailedStudent fs = new FailedStudent( rollNumber: 101, subjectCode: "BS101", subjectName: "Java Programming");
4
5      System.out.println("Failed Student Details:");
6      System.out.println("Roll No: " + fs.getRollNumber());
7      System.out.println("Subject Code: " + fs.getSubjectCode());
8      System.out.println("Subject Name: " + fs.getSubjectName());
9      System.out.println();
10
11     PromotedStudent ps = new PromotedStudent( rollNumber: 102, subjectCode: "BS102", subjectName: "Data Structures");
12
13     System.out.println("Promoted Student Details (Before Change):");
14     System.out.println("Roll No: " + ps.getRollNumber());
15     System.out.println("Subject Code: " + ps.getSubjectCode());
16     System.out.println("Subject Name: " + ps.getSubjectName());
17
18     ps.setSubjectName("Advanced Data Structures");
19
20     System.out.println("\nPromoted Student Details (After Change):");
21     System.out.println("Roll No: " + ps.getRollNumber());
22     System.out.println("Subject Code: " + ps.getSubjectCode());
23     System.out.println("Subject Name: " + ps.getSubjectName());
24 }
25 }

```

Output:-

```

C:\Users\LAPCOM\.jdk\openjdk-25\bin\java
Failed Student Details:
Roll No: 101
Subject Code: BS101
Subject Name: Java Programming

Promoted Student Details (Before Change):
Roll No: 102
Subject Code: BS102
Subject Name: Data Structures

Promoted Student Details (After Change):
Roll No: 102
Subject Code: BS102
Subject Name: Advanced Data Structures

Process finished with exit code 0

```

- Write a program that will calculate the below 4 formulas. Decide what to make mutable and what to make immutable and perform task operations. Formulas are:

Circumference of circle: $C = 2 \pi r$

Area of circle: $A = \pi r^2$

Volume of sphere: $V = \frac{4}{3} \pi r^3$

Surface area of sphere: $A = 4 \pi r^2$

(Hint: Value of pie would be constant and value of radius should be variant)

Code:-

```

1  final class MathConstant { no usages
2      public static final double PI = 3.14159;
3      private MathConstant() { no usages
4      }
5  }

MathConstant.java  ShapeCalculator.java  GeometryTest.java

1  class ShapeCalculator { no usages
2      private double radius; // Mutable variable 11 usages
3
4      public ShapeCalculator(double radius) { no usages
5          this.radius = radius;
6      }
7
8      public void setRadius(double radius) { no usages
9          this.radius = radius;
10     }
11
12     public double getRadius() { no usages
13         return radius;
14     }
15
16     public double circumferenceOfCircle() { no usages
17         return 2 * MathConstant.PI * radius;
18     }
19
20     public double areaOfCircle() { no usages
21         return MathConstant.PI * radius * radius;
22     }
23
24     public double volumeOfSphere() { no usages
25         return (4.0 / 3.0) * MathConstant.PI * radius * radius * radius;
26     }
27
28     public double surfaceAreaOfSphere() { no usages
29         return 4 * MathConstant.PI * radius * radius;
30     }
31 }

MathConstant.java  ShapeCalculator.java  GeometryTest.java

1  public class GeometryTest {
2      public static void main(String[] args) {
3          ShapeCalculator sc = new ShapeCalculator( radius: 7);
4          System.out.println("Radius: " + sc.getRadius());
5          System.out.println("Circumference of Circle: " + sc.circumferenceOfCircle());
6          System.out.println("Area of Circle: " + sc.areaOfCircle());
7          System.out.println("Volume of Sphere: " + sc.volumeOfSphere());
8          System.out.println("Surface Area of Sphere: " + sc.surfaceAreaOfSphere());
9
10         sc.setRadius(10);
11
12         System.out.println("\nAfter Changing Radius:");
13         System.out.println("Radius: " + sc.getRadius());
14         System.out.println("Circumference of Circle: " + sc.circumferenceOfCircle());
15         System.out.println("Area of Circle: " + sc.areaOfCircle());
16         System.out.println("Volume of Sphere: " + sc.volumeOfSphere());
17         System.out.println("Surface Area of Sphere: " + sc.surfaceAreaOfSphere());
18     }
19 }

```

Output:-

```
C:\Users\LAPCOM\.jdk\openjdk-25\bin
Radius: 7.0
Circumference of Circle: 43.98226
Area of Circle: 153.93791
Volume of Sphere: 1436.7538266666666
Surface Area of Sphere: 615.75164

After Changing Radius:
Radius: 10.0
Circumference of Circle: 62.8318
Area of Circle: 314.159
Volume of Sphere: 4188.7866666666666
Surface Area of Sphere: 1256.636

Process finished with exit code 0
```